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A4SLR: An Agentic Artificial Intelligence-Assisted Systematic Literature Review Framework to Augment Evidence Synthesis for Health Economics and Outcomes Research and Health Technology Assessment

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ABSTRACT

Objectives: Systematic literature reviews (SLRs) are essential for synthesizing high-quality evidence in clinical research, health economics and outcomes research, and health technology assessments. However, the growing volume of published data has made SLRs time-consuming, labor-intensive, and costly. To address these challenges, we introduce A4SLR, an agentic artificial-intelligence-assisted SLR framework, which provides a flexible, extensible methodology for automating the entire SLR process-from initial query formulation to evidence synthesis-across various study fields.

Methods: A4SLR comprises 8 modules integrated with specialized artificial intelligence agents powered by large language models: search, I/E criteria deployment, abstract/full-text screening, text/table pre-processing, data extraction, assessment, risk of bias analysis, and report. We implemented and validated this framework using 2 use cases, non-small cell lung cancer and perinatal mood and anxiety disorders. Performance of the assessment was evaluated quantitatively and qualitatively.

Results: Our implementation demonstrated high accuracy in article screening (F1 scores: 0.917-0.977), risk of bias assessment (Cohen's k: 0.8442-0.9064), and data extraction (F-scores: 0.96-0.998), including patient characteristics, safety and efficacy outcomes, economic model parameters, and cost-effectiveness data. Notably, the text/table preprocessing agent yielded comprehensive coverage of data elements, particularly in the challenging tasks of accurately matching outcome values to their corresponding study arms.

Conclusions: Our findings highlight the potential of the A4SLR framework to transform the evidence synthesis process by addressing the limitations of manual SLRs, thereby enhancing health economics and outcomes research and health technology assessments. Designed as a scalable, user-centric, extensible approach, A4SLR provides a robust solution for generating comprehensive up-to-date evidence to support researchers and decision makers across diverse clinical and therapeutic areas.

Keywords: agentic-AI, article screening, automation, data extraction, large language models, systematic literature review.

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Highlights

- We present A4SLR, a comprehensive agentic artificial intelligence-assisted systematic literature review framework that integrates structured human oversight.
- The framework implementation uses specialized multiagents, operating across modules to perform complex tasks, such as text processing, tabular data extraction, and risk of bias analysis, while ensuring adaptability across research domains.
- Validation of the framework demonstrates its potential to enhance the quality, efficiency, and scalability of evidence synthesis for health technology assessments and health economics and outcomes research.

Introduction

Systematic literature reviews (SLRs) are structured analyses that address specific research questions by using predefined protocols, comprehensive literature searches, and rigorous quality evaluation[1,2] to ensure unbiased, reproducible, and evidence-based conclusions.[3,4] In drug development, SLRs provide crucial data on treatment efficacy and safety profiles, supporting regulatory submissions to agencies such as the US Food and Drug Administration or the European Medicines Agency.[5-7] In health economics and outcomes research (HEOR), SLRs help evaluate the economic impact of diseases and treatments, including healthcare resource utilization cost and quality-of-life impacts,[8] demonstrating the value of new therapies to payers and policymakers through comprehensive health technology assessments (HTAs).[9-11] With the increasing emphasis on evidence-based practice,[12] the use of real-world evidence for drug approval,[13] and the requirement for HTAs in drug pricing and reimbursement,[14] SLRs have become indispensable for synthesizing high-quality, up-to-date evidence.[15,16] However, conducting SLRs has become more challenging because of the growing volume of studies[17,18] and the significant time[19,20] and cost involved,[21] highlighting the need for more efficient approaches.

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Automation, particularly with advances in artificial intelligence (AI) and natural language processing, has shown great potential to enhance SLR,[20,22-25] especially in literature screening and data extraction.[26,27] The integration of large language models (LLMs) has further expedited each phase of the SLR process, although their implementation requires caution and human oversight.[28,29] HTA agencies, such as the National Institute for Health and Care Excellence in the United Kingdom, are beginning to issue guidance on how to incorporate the use of AI into SLRs that are being submitted.[30] Fleurence et al[31] have also reviewed opportunities, challenges, and policy considerations about generative AI for HTA.

LLMs have been used in screening relevant articles,[32-34] conducting quality assessment,[35,36] and automating data extraction, including Population, Intervention, Comparator, and Outcome (PICO) elements.[37-40] Studies by Reason et al,[41] and Wang et al,[42] have explored the application of LLMs in automated network meta-analyses and developed evidence-synthesis pipelines focusing on treatment outcomes. Despite these advancements, comprehensive systems that integrate the entire SLR process, including querying relevant articles, screening based on the user-defined PICO criteria, and data extractions, risk of bias analysis, and summarization, with consistently high performance across multiple domains, such as treatment safety and efficacy, and economic evaluation analyses are lacking. Most existing studies focus on individual modules of the SLR process or specific study types. Additionally, the challenge of extracting comprehensive data from complex tables, especially in economic evaluation studies, remains largely unaddressed.

To address these challenges, we introduce an A4SLR framework for LLM-assisted, human-AI interactive SLRs that integrates multiagentic AI approaches into critical SLR workflows. The framework provides a flexible architecture that enables domain-specific customization and uses adaptable data extraction methodologies to enhance precision and efficiency across different research domains. The main contributions of our work are as follows:

1. A comprehensive framework for human-agentic AI collaborative SLRs: our framework presents a conceptual architecture that integrates agentic AI techniques with human oversight throughout the SLR workflow. This framework establishes a methodological foundation for automating the SLR pipeline while maintaining rigorous human validation checkpoints.
2. A methodological approach to managing complex tasks: we address the critical challenge in SLR by developing generalizable methods to extract granular data and assess risk of bias from intricate textual and tabular formats in full-text articles. This approach significantly enhances the reliability and completeness of evidence synthesis across varied study domains and publication formats.
3. Extensible and domain-adaptable design: our framework uses multiple interoperable AI agents that can be configured for different HEOR and HTA applications-whether treatment efficacy, cost-effectiveness, or other study types-providing a clear, yet flexible structure tailored to each field's needs.

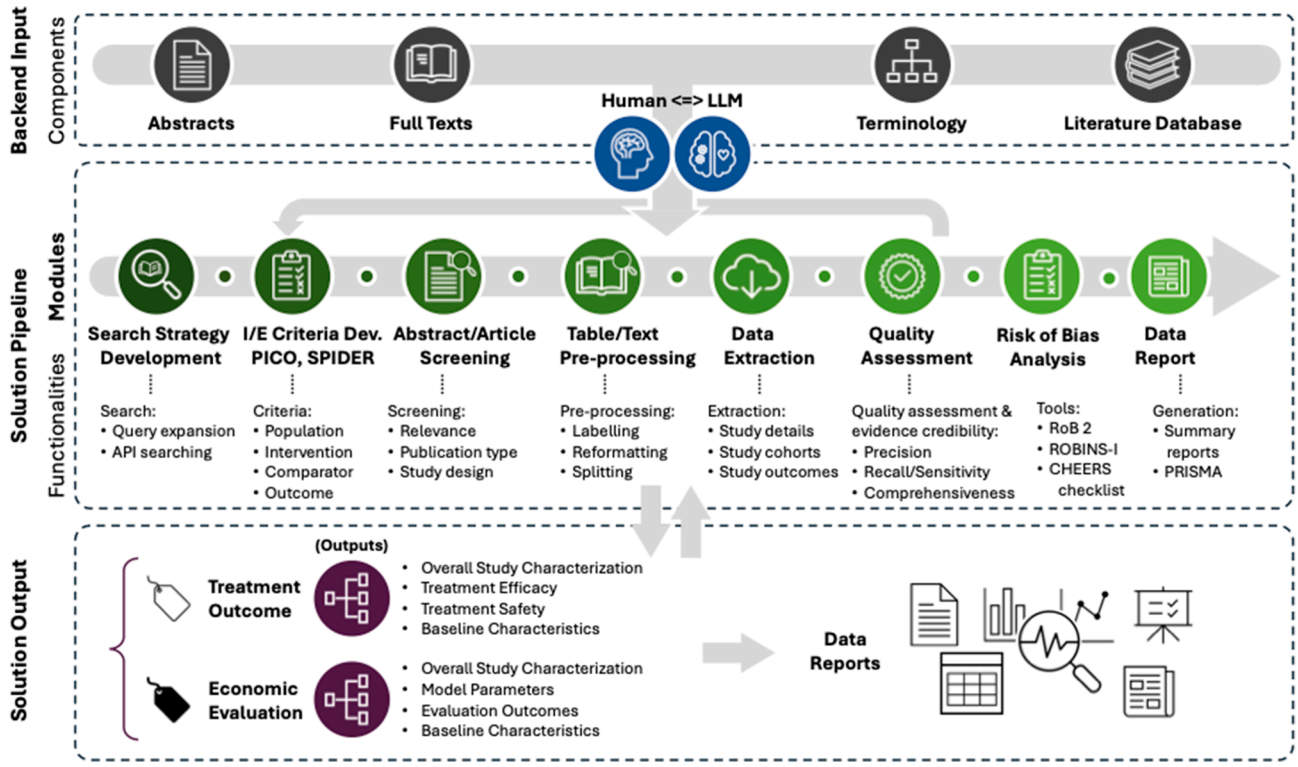
Methods

Framework Architecture

Our framework is designed as a modular, extensible architecture that integrates agentic AI capabilities with human expertise at critical decision points throughout the SLR process. Figure 1 illustrates the high-level conceptual architecture of the framework, showing the primary modules and their interactions. Figure 2 provides a detailed workflow view, depicting specific agent tasks and the strategic integration points between the AI agents and human experts. As demonstrated in Figure 2, each agentic module builds upon outputs from previous stages while enabling iterative refinement based on human feedback. The information flow follows a logical progression from query formulation through evidence synthesis, with human review checkpoints at critical decision points to ensure quality and relevance of outputs.

1. Design Principles: the framework is built around several core design principles.
   - 1.1. Modularity: Each agentic module addresses a specific function in the SLR process and can be implemented, improved, or replaced independently of other modules
   - 1.2. Human-AI collaboration: The framework incorporates strategic human oversight points to leverage AI efficiency while ensuring human judgement guides critical decisions.
   - 1.3. Domain adaptability: The architecture allows for customization to different research domains through configurable extraction templates and domain-specific screening criteria.
   - 1.4. Quality assurance: Multiple validation checkpoints are embedded throughout the process to ensure accuracy and completeness
   - 1.5. Extensibility: The framework can be extended with additional modules or enhanced capabilities as AI technologies evolve.
2. Framework Modules: the framework comprises 8 conceptual modules, each addressing a specific aspect of the SLR process. These modules were implemented using multiagentic AI approaches.
   - 2.1. Search: the function of the search module establishes the methodology for formulating effective search queries based on human-provided research objectives and expanding them to capture relevant literature comprehensively. Query formulation should balance specificity and sensitivity. Human researchers review and refine generated queries before execution. This module can be implemented using various approaches, from rule-based query expansion to AI-assisted query formulation and connecting to multiple literature search tools (eg, PubMed, Embase) via APIs. Query expansion through database techniques by mapping the search terms to medical subject headings, synonyms, and variants enhances retrieval completeness.
   - 2.2. Inclusion/Exclusion Deployment: this module establishes the methodology for developing, validating, and deploying screening criteria based on the PICO framework and additional relevance parameters. Different criteria may apply to the abstract and full-text screening phases. The module can be implemented using structured templates or AI-assisted criteria development tools. Human experts can define their own criteria or refine AI-generated criteria and approve final screening criteria before deployment.
   - 2.3. Screening: the screening module provides a systematic approach to evaluating research articles against defined criteria, with separate methodologies for abstract and full-text screening phases. The 2 independent agents' screening and checking agreement approach enhances reliability. Different implementation approaches can be

Figure 1. The architecture of A4SLR system.



API indicates Application Programming Interface; LLM, large language models; PICO, Population, Intervention, Comparator, and Outcome; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

used for abstract versus full-text screening based on complexity and resource constraints. Humans can review AI-assisted screening decisions, particularly discrepancies between dual AI-screeners, and make final determinations on ambiguous cases.

2.4. Preprocessor: this module establishes methodologies for preparing full texts and tables from "Relevant" articles for systematic data extraction, addressing challenges of document structure and content organization. The agent excludes sections such as Introduction, Discussion, and References to reduce false-positive data extraction. The resulting content is then split into manageable segments for further processing. Table preprocessing addresses complex hierarchical structures (see Appendix Fig. 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002). Domain-specific content categorization enhances extraction accuracy. Text preprocessing can be implemented through rule-based approaches (eg, token size segmentation), and table preprocessing may use specialized algorithms for structure normalization and content categorization. Humans may validate preprocessing results for complex documents, particularly those with complex table formats.

2.5. Extraction: the extraction module provides methodologies for systematically identifying and extracting relevant data elements from processed text and tables according to domain-specific templates. Extraction templates should be adaptable to different study domains, and hierarchical data relationships must be preserved during extraction. LLMs with different prompting strategies can be implemented. Humans refine extraction templates, validate extraction results, and refine prompts based on observed performance.

2.6. Assessment: this module is for evaluating the quality and completeness of extracted data and the overall SLR process. Both quantitative and qualitative assessment approaches are necessary. Error analysis should inform iterative framework improvements. Assessment can be implemented through automated metrics calculation. Human experts review assessment results, analyze error patterns, and inform refinements of the extraction agents.

2.7. Risk of Bias Analysis: this module establishes methodologies for systematically evaluating potential biases in included studies to determine whether a study's design, conduct, analysis, and reporting might have introduced systematic errors that could distort the true intervention effect. When integrated into the SLR process, this module strengthens the reliability of the conclusion by ensuring that evidence is generated from methodologically sound studies rather than potentially flawed ones. The risk of bias analysis should incorporate cross-validation techniques to minimize reviewer bias and should be tailored to specific research domains. Implementation typically occurs through standardized assessment tools (eg, Cochrane tools). The AI can function as a reviewer, whereas human experts serve as second reviewer to conduct independent bias assessments or resolve disagreements between evaluations of 2 AI reviewers as third reviewers. This dual-review process maintains methodological rigor while leveraging AI capabilities to enhance efficiency.

Figure 2. SLR tasks and the corresponding roles and interactions between the agentic AI and humans in the A4SLR framework.

SLR Agent	Agent Task	Agent Function		Human
1. Search	1.1. Research Objectives		←	Provide Research objectives
	1.2. Search Query Creation	Generate initial key search query based on the research objectives		
	1.3. Access to PubMed	Automatically access PubMed and collect abstracts using the query from step 1.2		
2. I/E deployer	2.1 . I/E criteria	Recommend I/E criteria based on the initial study objective and query strategy	↔	Define specific I/E criteria
	2.2 I/E criteria deployment	Process I/E criteria value and deploy it for screening		
3. Screening	3.1. Abstract Screening	Screen abstracts using I/E criteria by two independent AI Agents		
	3.2. Create Abstract Screening Results	Document screening result with AI recommendation (Relevant vs Irrelevant) and provide explanation	↔	Review of flagged abstracts (e.g., when AI Agents disagree) & finalize decision
	3.3. Full-text Collection	Downloading available full texts from PubMed for abstracts labeled "Relevant"		
	3.4. Full-text Screening	Perform full-text screening	↔	Review of flagged abstracts (e.g., when AI Agents disagree) & finalize decision
4. Pre-processor	4.1. Pre-processing Input Files	Convert files to text and separate text and tables.		
	4.2. Pre-processing Tables and Texts	Reorganize table format and tokenize content		
	4.3. Pre-processing Tables	Classification tables according to their content		
5. Extraction	5.1. Define Output	Recommend data elements based on I/E criteria	↔	Define output formats for data extraction
	5.2. Data Extraction	Extract data according to predefined output formats from		
	5.3. Data Integration	Merge data and remove duplicates		
	5.4. Create Extraction output	Document the data extraction results for each section	↔	Review and verify the extracted data
6. Assessment	6.1. Quality Performance Assessment	Access the performance through quantitative metrics against a human-annotated gold standard	↔	Create gold standard
7 Risk of Bias	7.1. Risk of Bias Analysis	Analyze the risk of bias based on the standard tool guidelines (e.g., RoB2, ROBINS-I)	↔	Review and verify
8. Report	7.1. Create Reports	Compile a comprehensive summary report based on PRISMA guideline	↔	Review and finalize the report

Red 1-head arrow: human input, blue 2-head arrow: human-AI interaction. AI indicates artificial intelligence; SLR, systematic literature review.

2.8. Report: this module is for synthesizing extracted data reports and Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) compliance reports. The extracted data report provides key information such as PMID, citation title, authors, year of publication, and data elements with corresponding values. Humans can review and use these reports for robust downstream data analysis. Additionally, the PRISMA compliance report ensures that the SLR process follows established standards for transparency and completeness. This helps evaluate whether the study has been conducted without bias and meets quality reporting standards.

Results

To validate our framework, we implemented it using agentic AI-based LLMs and evaluated on 2 distinct use cases: non-small cell lung cancer (NSCLC) for treatment outcomes and perinatal mood and anxiety disorders (PMAD) economic evaluations. This

Table 1. Data elements extracted from tables in oncology treatment studies and economic evaluation studies.

Study domain	Group	Elements
Oncology treatment outcome studies	Study details	"title," "author contact details," "year of publishing," "study design (eg, clinical-trial or real-world)," "registry number," "trial phase," "study duration," "masking" "patient segment," "description of interventions and control groups," "sample sizes for total and each group," "prior line of therapy," "treatment details (eg, cycle and dosage)," "data cutoff date,)" "duration of exposure," "duration of follow-up" and "study findings."
	Safety outcome	"cohort name," "cohort size (total and evaluated)," "evaluated cohort size," "outcome measure (eg, CRS, neutropenia etc)," "outcome measure grade (eg, all grades or grades 3 and 4)," "outcome measure type (eg, total, median, mean, average)," "outcome measure value," "measures of spread (eg, standard deviation, standard error, 95% confidence interval, others," "measures of spread ranges (eg, upper and lower values)," "*P* value" "comparator details," "odd or hazard ratio value w range," and "associated outcome for odd or hazard ratio"
	Efficacy outcome	"cohort name," "cohort size (total and evaluated)," "evaluated cohort size," "outcome measure (eg, overall response rate, complete response, progression free survival, etc)," "outcome measure type (eg, total, median, mean, average)," "outcome measure value," "measures of spread (eg, standard deviation, standard error, 95% confidence interval, others," "measures of spread ranges (eg, upper and lower values)," "*P* value," "comparator details," "odd or hazard ratio value w range," and "associated outcomes for odd or hazard ratio"
	Baseline characterization (both patient and tumor characteristics)	"cohort name," "cohort size," "characteristic element (eg, demographics, comorbidities, performance scores, such as ECOG, Karnofsky score, smoking status, treatment history, biomarkers, and histology), "*P* value," "comparator details," "odd or hazard ratio value w range," and "associated outcomes for odd or hazard ratio"
Economic evaluation studies	Study details	"title," "author contact details," "year of publishing," "study design (eg, model-based or trial-based)," "intervention type (eg, prevention effort, screening or treatment)," "description of interventions and control groups," "whether health disparities were mentioned," "sample sizes for total and each group," "county in which the study is conducted," "study period," "study group," "models used for simulation (eg, Markov model, decision tree, etc.)," "a summary of the link between cost and health benefit," "discount rate," "currency year," "funding source," and "the author's conflicts of interest information."
	Parameter	"type of parameter (eg, probability, utility, cost, etc.)," "study group," "name of characteristic," measure of central tendency (eg, mean, median, percentage, etc.," "measure of central," "measure of central tendency value," "measure of spread (eg, standard deviation (SD), standard error (SE), etc.)," "measures of spread (if others," "measure of spread value," "upper bound of measure of spread," "lower bound of measure of spread," and "source."
	Economic evaluation outcomes	"type of economic evaluation (eg, cost-effectiveness, cost-utility, etc)," "evaluation details (if available)," "study cohorts," "time horizon," "outcome type (eg, cost, quality-adjusted life year (QALY), incremental cost-effectiveness ratio (ICER), net present value (NPV), etc)," "outcome measure type (eg, total, mean, net, per life year, probability, etc)," "outcome measure value," "currency" and "source"
	Baseline characteristics	"demographics (eg, age, gender, race, and ethnicity)," "comorbidities," and "socioeconomic status"

implementation serves as one possible instantiation of the framework concepts and demonstrates its practicality and effectiveness.

1. Implementation Details:
   1.1. Screening and Data Extraction: in our implementation, we deployed agentic AI approaches across various modules, utilizing GPT-4 turbo and GPT-4o LLMs. These models were strategically selected for their different strengths in text processing and tabular data extraction. Building on our previous research on LLM-based tabular data extraction from conference abstracts,[38] we adopted agentic AI approaches to manage more complex tables in full-text articles across additional study domains. Appendix Figure 2 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002 provides an

Table 2. Performance metrics of abstract screening process for NSCLC and PMAD.

Total	Gold: relevant	TP	TN	FP	FN	Precision	Recall	F1-score	Specificity	Accuracy
NSCLC (*n* = 865)	341	335	514	10	6	0.971	0.982	0.977	0.971	0.982
PMAD (*n* = 596)	66	61	524	6	5	0.910	0.924	0.917	0.924	0.981

Note. Precision = TP/ (TP + FP), Recall (sensitivity) =TP/(TP + FN), F1-Score = 2 × ([Precision × Recall]/[Precision + Recall]), Specificity = TN/(TN + FP), and Accuracy = TP + TN/(TP + TN + FP + FN).
FN indicates false negative; FP, false positive; NSCLC, non-small cell lung cancer; PMAD, perinatal mood and anxiety disorders; TN, true negative; TP, true positive.

example of a preprocessed table input file from a study on NSCLC treatment outcomes.[43] Our approach incorporated several key processing steps: (1) the agent segments long tables into smaller parts to enhance data extraction accuracy. (2) The agent categorizes tables by study types and content. (3) For treatment outcome studies, tables were labeled as either "safety outcome," "efficacy outcome," or "baseline characteristics/associated risk factors." (4) For economic evaluation studies, tables were labeled as "parameters (including baseline characteristics tables if present)" or "evaluation outcomes." Table 1 provides a detailed list of data elements extracted for each study type and group.

1.2. Quality Assessment: we quantitatively evaluated the agent's performance on both screening and data-extraction tasks by comparing its outputs with a human-annotated gold standard. Standard classification metrics were computed for screening and data extraction. Precision is calculated as the ratio of correctly predicted positive entities to the total predicted positive entities (Precision = True Positive/(True Positive + False Positive)). Recall, also known as sensitivity, is calculated as the ratio of correctly predicted positive entities to all actual positive entities (Recall = True Positive/(True Positive + False Negative)). The F1-score is the harmonic mean of precision and recall and is calculated using the formula: F1-score = 2 × ((Precision × Recall)/(Precision + Recall)).

1.3. Risk of Bias Analysis: we applied Cochrane Risk Of Bias In Non-randomized Studies-of Interventions (ROBINS-I V2)[44] to evaluate nonrandomized intervention studies. ROBINS-I V2 includes structured questions and decision algorithms across 7 bias domains—(1) confounding, (2) classification of interventions, (3) participant selection, (4) deviations from intended interventions, (5) missing data, (6) outcome measurement, and (7) selection of reported results. These structured question prompts, and domain-specific guidance were embedded into the AI model along with the accompanying contextual descriptions. The model then applied the tool algorithms to assign each domain—and the overall study—a bias rating of low, moderate, severe, or critical based on its responses. To validate performance, we randomly selected 135 questions across all domains from 5 articles, compared the model's answers with the human-curated gold standard, and calculated Cohen's κ for each domain.

1.4. Reporting: for the PRISMA compliance report, the agent recorded the total number of articles excluded under each exclusion criterion. We then randomly sampled 50 excluded articles per criterion and assessed whether the AI's exclusion decisions matched human reviewer labels, ensuring transparency.

2. Implementation Performance

2.1. Searching and Screening: AI-assisted screening allows us to use recall-focused, broader search terms compared with manual searching strategy work because of the significant reduction in the initial screening time. In both NSCLC and PMAD, the AI-assisted search strategy successfully retrieved all articles captured by human queries. The implemented system took 2 hours and 40 minutes for the screening of eligibility using the PICO criteria (see Appendix Fig. 3 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002 for NSCLC example) from the corpus of 2135 NSCLC and 1605 PMAD studies searched based on queries. Among them, randomly selected 865 NSCLC studies and 596 PMAD studies were evaluated. It achieved F1-scores of 0.917 to 0.989 across both use cases (refer Table 2 for details), with particularly strong performance in full-text screening (100% agreement with human reviewers).

2.2. Text and Table Data Extraction: the implemented system took 10 to 30 mins per article to extract data and achieved F1-scores of 0.961 to 0.998 for text data element extraction, including complex statistical values and cohort-specific outcomes. Detailed performance scores of GPT-4o and GPT 4 turbo in identifying individual features from text are provided in Figure 3 and Appendix Table 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002. We found that GPT4o performs superiorly in study cohort identification and corresponding outcome value extraction.

For table data extraction, we used the GPT-4 turbo model, leveraging its larger token size limit to process complex tables efficiently. The implementation also effectively extracted hierarchical data from complex tables, demonstrating the effectiveness of the preprocessing methodology (illustrated in Appendix Fig. 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002). Detailed performance scores for tabular data extraction across 18 NSCLC articles and 18 PMAD articles are presented in Table 3 (ranging from 0.961 to 0.99 of F1 scores). Each article contains approximately 3 to 5 tables. Notably, GPT-4 turbo demonstrated superior performance in extracting data from tables (F1 scores ranging from 0.961 to 0.998, with an average of 0.984) compared with text extraction (F1 scores ranging from 0.864 to 1, with an average of 0.958) in NSCLC treatment outcome studies. This contributed to improved performance in accurately identifying study cohorts and subgroups when handling structured data formats.

2.3. Risk of bias analysis: for each article, the table reports "Domain number and description, Question ID and category, Question, LLM-response and LLM-rationale, and Risk of Bias"

Figure 3. Comparison of precision, recall, and F1 scores between GTP-4 turbo and GTP-4o across various data elements in text-data extraction.



Table 3. Performance metrics of GPT-4 turbo for extracting table-data from NSCLC and PMAD articles.

Study type	Data element group	Precision	Recall	F1-score
Treatment outcome study (NSCLC, *N* = 18)	Study details	0.994	0.996	0.995
	Safety outcomes	0.997	0.998	0.997
	Efficacy outcomes	0.999	0.997	0.998
	Baseline characteristics	0.983	0.946	0.961
	Average	0.993	0.988	0.984
Economic evaluation study (PMAD, *N* = 18)	Study details	0.998	0.989	0.993
	Model parameters	1	0.996	0.998
	Evaluation outcomes	1	0.997	0.998
	Baseline characteristics	0.986	0.985	0.984
	Average	0.996	0.992	0.993

NSCLC indicates non-small cell lung cancer; PMAD, perinatal mood and anxiety disorders.

rating at the domain level and the overall level (each rated as low, moderate, serious, or critical). Appendix Table 2 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002 presents Cohen's k scores comparing the AI agent's judgements with human-curated benchmark across 7 domains. The overall Cohen's k score was 0.8819 (range: 0.8442-0.9064).

2.4. Reporting: Appendix Table 3 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002 shows how often the AI's eligibility decisions matched those of human reviewers across 7 criteria (3 population, 3 intervention/comparator, and 1 outcome rule). Of the 50 articles evaluated per criterion, the AI agent perfectly agreed with the human labels on 5 criteria (100 % agreement for Population criterion 1 and 3, Intervention/Comparator criterion 1 and 3, and Outcome criterion 1) and achieved 98 % agreement on the remaining 2 criteria (Population criterion 2 and Intervention/Comparator criterion 2).

2.5. Qualitative Error Analysis: the system utilized both GPT-4 turbo and GPT-4o models for extracting data from text, whereas GPT-4 turbo model was used for extracting data from tables. Error analysis of text-based data extraction revealed key differences, as summarized in Appendix Table 4 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002. For "Study Details," GPT-4 turbo frequently

misses essential information, such as the "Value" of study design specifics (eg, "randomized") and demographic details, such as sex ratios and prior treatments. In the "Study Outcome" category, although generally effective, GPT-4 turbo struggled with accurately specifying study cohorts and subgroups, particularly in multitrial comparison studies. In terms of data elements and statistical values, GPT-4 turbo often misses the "Value" of important elements, such as "Study duration" and "Data cutoff" dates, and struggles to separate statistical values, such as hazard ratios (HR), odds ratios, 95% confidence intervals (CI), ranges, and *P* values. This often results in marking values as "NA" or providing incorrect information. GPT-4o demonstrated higher accuracy in capturing these elements and distinguishing between various statistical metrics, although it occasionally fails to separate specific metrics clearly. Table data extraction challenges included managing multiple hierarchical levels (groups, subgroups, and subsubgroups) associated with a single value and their inconsistent placement (see Appendix Fig. 4 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.08.002). We also identified some errors propagated from the "pdf-to-text" conversion step, such as symbols being improperly translated (eg, ">=" converted to ">"), affecting the accuracy of extracted outcomes

Discussion

We introduce A4SLR, a conceptual framework that merges LLM-powered, multiagentic AU with human interaction across the entire process of SLR, including searching, I/E criteria, screening, data extraction, quality assessment, risk of bias analysis, and final reporting. To demonstrate its versatility, we applied A4SLR in 2 different study domain use cases: NSCLC for treatment effectiveness safety outcomes across clinical trials and real-world studies, and PMAD for economic model parameterization and cost-effectiveness analysis. Validation of our implementation showed high accuracy and revealed the critical importance of the preprocessing module and human oversight.

Implementation of our framework successfully retrieved all articles captured by expert queries and achieved high performance in abstract and full-text screening across both treatment outcome and economic evaluation study domains. In previous screening tasks utilizing LLMs, the sensitivity of abstract screening ranged from 81.1% to 96.5% in Tran et al[34] study and 76% to 91% in Guo et al[32] study. Guo et al[32] discussed the challenges of lower sensitivity for including articles, reporting 76% as a limitation. Our study achieved the sensitivity of 98.2% and the specificity of 97.1% for NSCLC studies. For PMAD, both sensitivity and specificity were 92.4%. By dividing the literature screening process into 2 distinct phases—abstract screening and full-text screening—the system effectively addresses the inherent differences between abstracts and full texts in terms of scope and detail. Abstracts provide a concise overview, summarizing the research question, methods, key findings, and conclusions, whereas full texts offer a more detailed understanding of the research methodology, findings, and implications. Our system allows for broader criteria for abstract screening to capture a wide array of relevant studies and more specific criteria for full-text screening to ensure precision.

The implemented system also demonstrated granularity and depth in data extraction, successfully identifying a wide range of treatment-related outcomes, economic evaluation model parameters, evaluation outcomes, and corresponding values. Previous studies, such as Gartlehner et al,[39] which achieved 96.3% accuracy in extracting 16 different data elements compared with human reviewers, and Reason et al,[41] which reached 99% accuracy in replicating data extraction from 4 network meta-analyses, highlight the potential of AI in this domain. Consistent with these findings, our implementation demonstrated high performance in extracting 40 different data elements each in treatment outcome and economic evaluation studies, achieving F1 scores ranging from 0.961 to 0.998 against human-curated gold standards. These results indicate that LLMs can achieve high performance even in extracting very complicated data elements when prompts are specifically tailored.

Our comparison between GPT-4 turbo and GPT-4o models highlighted that different models showed varying strengths, with GPT-4o excelling at extracting complex relationships from text and GPT-4 turbo performing better on tabular data because of its larger context window. Notably, GPT-4 turbo demonstrated superior performance in table data extraction (F1 scores ranging from 0.961 to 0.998, average 0.984) compared with text extraction (F1 scores ranging from 0.864 to 1, average 0.958), improving the accurate identification of study cohorts and subgroups in structured data formats.

Our study has several notable strengths. Our multiagentic AI SLR framework addresses limitations of existing approaches to automated SLRs. Unlike approaches focusing on specific SLR tasks (eg, screening only), our holistic framework spans the entire SLR process—searching, I/E criteria, screening, data extraction, quality assessment, risk of bias analysis, and reporting. The framework's modular design enables expansion to different research domains, unlike methods optimized for single domains. Additionally, our framework explicitly incorporates human oversight at strategic points rather than fully automated approaches that may lack necessary validation.

Despite its strengths, we also acknowledge several limitations. Each domain-specific adaptation needs validation before deployment in real-world settings. Our current validation covered only a single use case per study domain (treatment outcome and economic evaluation), which limits its generalizability. To enhance the framework's utility, future work should include systematic validation across multiple research domains to demonstrate broader applicability and robustness. Moreover, the implementation and validation of various risk of bias analysis tools, which are appropriate for different study designs, is necessary. For example, Cochrane Risk of Bias 2 (RoB 2) for randomized controlled trial studies and Consolidated Health Economic Evaluation Reporting Standards checklist for economic evaluation studies in addition to the current ROBINS-I V2 for nonrandomized studies that compare health interventions. This approach can ensure domain-specific assessment criteria across a diverse study corpus. Furthermore, our framework can be extended by integrating additional modules such as image and figure data extraction, and implementation of comprehensive meta-analysis and visualization modules. Lastly, our implementation of reporting automation focused on the extraction and formatting of PRISMA eligibility elements (for the PRISMA flow diagram) and the structured listing of included studies. Although we did not evaluate the AI's ability to generate complete narrative reports, detailed subheadings, or discussion text, these represent exciting opportunities for future development. Enhancing human-AI collaboration in full dossier drafting may ensure that report layout and emphasis meet specific journal or stakeholder requirements. Such improvements would expand the framework's functionality and increase its value for evidence synthesis in HEOR.

Conclusions

We provided a comprehensive and reliable framework for conducting SLRs by integrating multiagentic AI methodologies

with strategic human oversight. Our implementation demonstrates the framework's potential to significantly enhance the efficiency and reliability of evidence synthesis with high accuracy in screening, data extraction, and risk of bias analysis across different research domains. The framework's modular architecture enables adaptation to diverse research contexts, and its extensibility allows evolution alongside advancements in AI technologies. By providing a conceptual foundation for multi-agentic AI integration in SLR, our framework contributes to timely and reliable evidence generation for advancing HTAs and HEOR.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section. Drs Dawoud and Shi are editors for *Value in Health* and had no role in the peer-review process of this article.

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